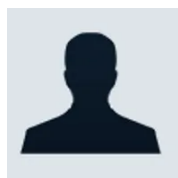


ZDNET[Home](#) / [Tech](#) / [Security](#)

JPEG worm breaks new ground

Dubbed the first 'JPEG infector,' a new multipart virus uses image files to stage an attack on PCs. Security experts say the real threat lies not in the worm--but in its innovations.



Written by **Robert Lemos**, Contributor

June 13, 2002 at 11:50 a.m. PT



CLICK TO UNMUTE





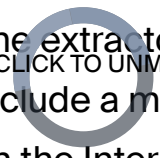
Antivirus companies warned on Thursday of a new virus that communicates through digital images, but security experts aren't sure how much of a threat this latest evolutionary branch of malicious code poses.

Dubbed the first "JPEG infector" by security company Network Associates, the W32/Perrun virus has two parts: infected JPEG images that contain the virus's payload and a viral program that extracts the code from the images and infects other JPEGs on the system as they are opened.

Because PCs have to be infected by the extractor virus before any code hidden in image files can affect them, the program is more a computer-science curiosity than a threat, said Vincent Gullotto, vice president of Network Associates' antivirus emergency response team.

"We are not saying that this is a problem," Gullotto said. "We gave it a low risk, but we haven't seen anything like this before." A digital image carrying code for W32/Perrun is easy to spot, he said, because the image is corrupted by the new code.

PC users should note that they can't be infected by opening a JPEG image. Rather, a virus on an infected computer copies code into a digital image and waits for the JPEG to get passed along to other infected systems. The virus on those systems will read the code fragment in the JPEG image and follow the instructions. Users who haven't been infected by the extractor virus can open an infected digital image and nothing will happen.

The extractor file only infects computers running Microsoft Windows and doesn't include a mass-mailing component. And, in fact, the virus has not been released on the Internet, but has been sent only to major antivirus companies by the



creator of the code.



**Get up-to-the-minute tech
news, security alerts, and more**

Sign up for ZDNET's daily News Alerts

Email address

Sign Me Up

By clicking Sign Me Up, you confirm you are 16+ and agree to our [Terms of Use](#) and [Privacy Policy](#).

However, the code has opened up a debate among antivirus researchers as to whether viruses with multiple parts could represent a new threat to PC users.

With some rather simple improvements, the virus could pose a threat, Gullotto said.

One obvious modification, which has already been discussed among the virus community, is using steganography--a technique to hide data in pictures--to allow such programs to embed code in images without corrupting the picture.

An upgradable virus is not a new event in the virus world. Hybris, a worm that slowly infected a large number of computers on the Internet last year, could be upgraded with encrypted plug-ins that were posted to Usenet, security experts have said.

Researchers have long worried about the evolving technology in viruses, and the latest critter to climb out of the Internet shows that the arms race with virus writers hasn't slowed.



But for Gullotto, the real lesson is one of foresight.

"People should start becoming more leery of JPEG files," he said. "If there is a chance that we can get ahead of the virus curve in educating the users, we should."

 **Editorial standards**

[show comments](#)

**we equip you to harness the power of
disruptive innovation, at work and at
home.**



topics

galleries

videos

do not sell or share my personal information

about ZDNET

meet the team

sitemap

reprint policy

join | log in

newsletters

licensing

accessibility

© 2026 ZDNET, A Ziff Davis company. All rights reserved. **Privacy Policy** || **Your Privacy Rights** | **Advertise** | **Terms of Use**

